DPEN022: Enabling Mathematics B
Solutions to Class Questions Week 3

Week 3A

Key Idea – Derivative of Powers of x

1. (a) $f(x)=7$

$f'(x)=0$

(b) $y=5x$

$\frac{dy}{dx}=5$

(c) $f(x)=-\frac{1}{3}x$

$f'(x)=-\frac{1}{3}$

(d) $f(x)=\frac{x}{6}=\frac{1}{6}x$

$f'(x)=\frac{1}{6}$

(e) $y=x=1x$

$\frac{dy}{dx}=1$

(f) $f(x)=x^2$

$f'(x)=2x^{2-1}$

$f'(x)=2x$

(g) $f(x)=x^3$

$f'(x)=3x^{3-1}$

$=3x^2$

(h) $y=\frac{2}{3}x^6$

$\frac{dy}{dx}=\frac{2}{3}\times 6x^{6-1}$

$=4x^5$

(i) $f(x)=7x^5$

$f'(x)=7\times 5x^{5-1}$

$=35x^4$

(j) $f(x)=\frac{2x^2}{5}=\frac{2}{5}x^2$

$f'(x)=\frac{2}{5}\times 2x^{2-1}$

$=\frac{4}{5}x$ or $\frac{4x}{5}$

(k) $y=-x^4$

$\frac{dy}{dx}=-1\times 4x^{4-1}$

$=-4x^3$

(l) $f(x)=\frac{1}{2}x^3$

$f'(x)=\frac{1}{2}\times 3x^{3-1}$

$=\frac{3}{2}x^2$ OR $\frac{3x^2}{2}$

(m) $y=\frac{2}{3}x^{-6}$

$\frac{dy}{dx}=\frac{2}{3}\times(-6)x^{-6-1}$

$=-4x^{-7}$

$=\frac{-4}{x^7}$

(n) $y=-2x^{-4}$

$\frac{dy}{dx}=-2\times -4x^{-4-1}$

$=8x^{-5}$

$=\frac{8}{x^5}$

(o) $f(x)=5x^{-3}$

$f'(x)=5\times -3x^{-3-1}$

$=-15x^{-4}$

(p) $f(x)=x^{-2}$

$f'(x)=-2x^{-2-1}$

$=-2x^{-3}$

$=-\frac{2}{x^3}$

(q) $y=\frac{-3}{x^5}=-3x^{-5}$

$\frac{dy}{dx}=-3\times(-5)x^{-5-1}$

$=15x^{-6}$

$=\frac{15}{x^6}$

(r) $y=\frac{-2}{3x^4}=\frac{2}{3}x^{-4}$

$\frac{dy}{dx}=\frac{2}{3}\times(-4)x^{-4-1}$

$=\frac{-8}{3}x^{-5}$

$=\frac{-8}{3x^5}$

2.

(a) $y = x^{\frac{1}{4}}$

$y' = \frac{1}{4}x^{-\frac{3}{4}}$

$= \frac{1}{4x^{\frac{3}{4}}}$ or $= \frac{1}{4\sqrt[4]{x^3}}$

(b) $f(x) = 2x^{0.4}$

$f'(x) = 2 \times 0.4x^{-0.6}$

$= 0.8x^{-0.6}$

(c) $y = -3x^{\frac{2}{3}}$

$\frac{dy}{dx} = -3 \times \frac{2}{3}x^{-\frac{1}{3}}$

$= -2x^{-\frac{1}{3}}$

$= \frac{-2}{x^{\frac{1}{3}}}$ or $= \frac{-2}{\sqrt[3]{x}}$

(d) $y' = \frac{1}{16}x^{\frac{-3}{4}} = \frac{x^{\frac{-3}{4}}}{16}$ negative index form

$y' = \frac{1}{16x^{\frac{3}{4}}}$ positive index form

$y' = \frac{1}{16\sqrt[4]{x^3}}$ positive surd form

(e) $y' = 8 \times \frac{3}{4}x^{\frac{-1}{4}} = 6x^{\frac{-1}{4}}$

$= \frac{6}{x^{\frac{-1}{4}}} = \frac{6}{\sqrt[4]{x}}$

(f) $y' = \frac{-2}{3} \times \frac{2}{3}x^{\frac{-1}{3}} = \frac{-4}{9}x^{\frac{-1}{3}}$

$= \frac{-4}{9x^{\frac{1}{3}}} = \frac{-4}{9\sqrt[3]{x}}$

(g) $\frac{dy}{dx} = \frac{1}{2}x^{-1/2} = \frac{1}{2x^{1/2}}$

$= \frac{1}{2\sqrt{x}}$

(h) $y = 4\sqrt{x^3} = 4x^{3/2}$

$y' = 4 \times \frac{3}{2}x^{1/2} = 6x^{1/2} = 6\sqrt{x}$

(i) $f(x) = -7\sqrt[4]{x} = -7x^{\frac{1}{4}}$

$f'(x) = -7x^{\frac{1}{4}} = -7 \times \frac{1}{4}x^{\frac{-3}{4}}$

$= \frac{-7}{4x^{\frac{3}{4}}} = \frac{-7}{4\sqrt[4]{x^3}}$

(j) $y = \frac{-2}{3x^{0.25}} = \frac{-2x^{-0.25}}{3}$

$y' = \frac{-2}{3} \times (-0.25)x^{-1.25} = \frac{-1}{6}x^{-1.25} = \frac{-1}{6x^{1.25}}$

(k) $y' = -2 \times \left(\frac{-3}{2}\right)x^{-\frac{5}{2}} = 3x^{-\frac{5}{2}}$

$= \frac{3}{x^{-\frac{5}{2}}} = \frac{3}{\sqrt{x^5}}$

(l) $y = \frac{-1}{x} = -1x^{-1}$

$y' = -1 \times (-1)x^{-2} = x^{-2} = \frac{1}{x^2}$

(m) $f(x) = \frac{1}{\sqrt{x}} = x^{-\frac{1}{2}}$

$f'(x) = \frac{-1}{2}x^{-\frac{3}{2}} = \frac{-1}{2x^{\frac{3}{2}}} = \frac{-1}{2\sqrt{x^3}}$

(n) $y = \frac{-2}{3\sqrt{x}} = \frac{1}{3}x^{-\frac{1}{2}}$

$y' = \frac{1}{3} \times \left(\frac{-1}{2}\right)x^{-\frac{3}{2}} = \frac{-1}{6}x^{-\frac{3}{2}}$

$= \frac{-x^{-\frac{3}{2}}}{6} = \frac{-1}{6x^{\frac{3}{2}}} = \frac{-1}{6\sqrt{x^3}}$

(o) $f(x) = \frac{1}{5\sqrt{x^3}} = \frac{1}{5}x^{-\frac{3}{2}}$

$f'(x) = \frac{1}{5} \times \left(\frac{-3}{2}\right)x^{-\frac{5}{2}} = \frac{-3}{10}x^{-\frac{5}{2}}$

$= \frac{-3x^{-\frac{5}{2}}}{10} = \frac{-3}{10x^{\frac{5}{2}}} = \frac{-3}{10\sqrt{x^5}}$

(p) $f(x) = x\sqrt{x} = x^{1+1/2} = x^{3/2}$

$f'(x) = \frac{3}{2}x^{1/2} = \frac{3\sqrt{x}}{2}$

(q) $f(x)=\dfrac{\sqrt{x}}{x}=x^{1/2-1}=x^{-1/2}$

$f'(x)=\dfrac{-1}{2}x^{-3/2}=\dfrac{-1}{2x^{3/2}}=\dfrac{-1}{2\sqrt{x^3}}$

(r) $y=\dfrac{x^2}{\sqrt{x}}=x^{2-\frac{1}{2}}=x^{\frac{3}{2}}$

$\dfrac{dy}{dx}=\dfrac{3}{2}x^{\frac{1}{2}}=\dfrac{3x^{\frac{1}{2}}}{2}=\dfrac{3\sqrt{x}}{2}$

Key Idea – Derivative of Powers of *x* using Rule 1 and 2

1. (a) $y'=0+12x^3$
$=12x^3$

(b) $\dfrac{dy}{dx}=2x-3+0$
$=2x-3$

(c) $f'(x)=-6x^{-4}+0$
$=\dfrac{-6}{x^4}$

(d) $y'=2+2x^3-0$
$y'=2+2x^3$

(e) $f(x)=\dfrac{x^2}{4}+3x-12.5$
$f'(x)=\dfrac{x}{2}+3-0$
$f'(x)=\dfrac{x}{2}+3$

(f) $y'=\dfrac{x^2}{2}+\dfrac{2}{3}$

(g) $y=2x^3-\sqrt[4]{x}+\dfrac{2}{x^2}+1$
$=2x^3-x^{1/4}+2x^{-2}+1$
$y'=6x^2-\dfrac{1}{4}x^{-3/4}-4x^{-3}$
$=6x^2-\dfrac{1}{4x^{3/4}}-\dfrac{4}{x^3}$

(h) $y=\sqrt[5]{x}+x^{-3/4}$
$=x^{1/5}+x^{-3/4}$
$y'=\dfrac{1}{5}x^{-4/5}-\dfrac{3}{4}x^{-7/4}$
$=\dfrac{1}{5x^{4/5}}-\dfrac{3}{4x^{7/4}}$

(i) $y=3x+\dfrac{3x^2}{\sqrt{x}}=3x+3x^2x^{1/2}2.$
$=3x+3x^{2-1/2}$
$=3x+3x^{3/2}$
$y'=3+3\times\dfrac{3}{2}x^{1/2}$
$=3+\dfrac{9}{2}x^{1/2}$

2. (a) $y=x(2x+1)$
$y=2x^2+x$
$\dfrac{dy}{dx}=2\times 2x^{2-1}+1$
$=4x+1$

(b) $y=(1+x^2)(x^2+3x)$
$=x^4+3x^3+2x+3$
$y'=4x^3+9x^2+2$

(c) $y=\dfrac{1}{5}\left(7x^3-\dfrac{1}{2}x^2+2x-3\right)$
$y'=\dfrac{1}{5}(21x^2-x+2)$
$=\dfrac{21x^2}{5}-\dfrac{1}{5}x+\dfrac{2}{5}$

(d) $f(x)=(x+4)(x-4)$
$f(x)=x^2-4^2$
$=x^2-16$
$f'(x)=2x^{2-1}$
$=2x$

(e) $f(x)=(3x-1)^2$
$=(3x)^2-2(3x)+1^2$
$=9x^2-6x+1$
$f'(x)=9\times 2x^{2-1}-6$
$=18x-6$

(f) $f(x)=(1+x^3)^2$
$=1+2x^3+x^6$
$f'(x)=6x^2+6x^5$

(g) $f(x)=\dfrac{2x^2+3x^3}{x^2}$
$=2+3x$
$f'(x)=3$

(h) $f(x)=\dfrac{8x^2+6x+1}{2x}$
$=4x+3+\dfrac{1}{2}x^{-1}$
$f'(x)=4-\dfrac{1}{2}x^{-2}=4-\dfrac{1}{2x^2}$

(i) $y=\dfrac{2x^3+5x}{x}=\dfrac{x(2x^2+5)}{x}$
$y=2x^2+5$
$\dfrac{dy}{dx}=2\times 2x^{2-1}=4x$

<u>Week 3B</u>

Key Ideas – Basic Application of Derivatives

1. $f(x)=x^2+6x$ Now solve $f'(x)=2x+6=0$

$$f'(x)=2x+6$$

$$2x+6=0$$
$$2x=-6$$
$$x=-3$$

$\therefore x=-3$ when $f'(x)=0$

2. $y=x^2-1$

$$y'=2x$$
$$y'(3)=6$$

Therefore gradient = 6 when x = 3.

3. $y=x^2-1$

$$y'=2x$$

$$2x=-8$$
$$x=-4$$

$$y(-4)=(-4)^2-1$$
$$=15$$

$\therefore$ coordinates are (–4, 15)

4. Differentiate:

$$y=15x-2x^3$$
$$\frac{dy}{dx}=15-6x^2$$

Substitute: $\frac{dy}{dx}=15-6(1)^2$

$$=9$$

Therefore, the gradient of the tangent to the curve at the point (1, 13) is 9.

5.

Differentiate:

$$f(x)=3x^2-2x+5$$
$$f'(x)=6x-2$$

Substitute: $f'(-2)=6(-2)-2$

$$=-14$$

Therefore, the gradient of the function at $x=-2$ is -14.

6. Differentiate:

$$y=2x^2+3x$$
$$\frac{dy}{dx}=4x+3$$

Solve for: $\frac{dy}{dx}=15$

$$4x+3=15$$
$$4x=12$$
$$x=3$$

Use the equation of the curve to find the y-coordinate when $x=3$:

$$y=2(3)^2+3(3)$$
$$=27$$

Therefore, (3,27) is the point on the curve where the gradient is equal to 15.

7.

Differentiate:

$$f(x)=\frac{1}{x}=x^{-1}$$
$$f'(x)=-x^{-2}$$
$$=-\frac{1}{x^2}$$

Substitute: $f'(2)=-\frac{1}{2^2}$

$$=-\frac{1}{4}$$

8.

Differentiate:

$$f(x) = 12x - x^3$$
$$f'(x) = 12 - 3x^2$$

Solve for:

$$\begin{aligned} f'(x) &= 0 \\ 12 - 3x^2 &= 0 \\ 3x^2 &= 12 \\ x^2 &= 4 \\ x &= \pm\sqrt{4} \\ &= \pm 2 \end{aligned}$$

Therefore, $x = 2, -2$ are the values of x where $f'(x) = 0$ (the gradient is zero).

Key Idea – Higher Order Derivatives

1.
$$\begin{aligned} f'(x) &= 7x^6 - 10x^4 + 4x^3 - 1 \\ f''(x) &= 42x^5 - 40x^3 + 12x^2 \\ f'''(x) &= 210x^4 - 120x^2 + 24x \\ f^{(4)}(x) &= 840x^3 - 240x + 24 \end{aligned}$$

2. Find $\dfrac{d^2y}{dx^2}$, if $y = x^3 - \dfrac{5}{x}$

3.
$$\begin{aligned} f(x) &= x^4 - x^3 + 2x^2 - 5x - 1 \\ f'(x) &= 4x^3 - 3x^2 + 4x - 5 \\ f''(x) &= 12x^2 - 6x + 4 \\ f''(-1) &= 12 + 6 + 4 \\ &= 22 \end{aligned}$$

4.
$$\begin{aligned} f(t) &= t^4 - 2t + 1 \\ f'(t) &= 4t^3 - 2 \\ f''(t) &= 12t^2 \\ f''(-2) &= 48 \end{aligned}$$

5.
$$\begin{aligned} y &= x^5 - 3x^4 + x \\ \frac{dy}{dx} &= 5x^4 - 12x^3 + 1 \\ \frac{d^2y}{dx^2} &= 20x^3 - 36x^2 \\ \frac{d^3y}{dx^3} &= 60x^2 - 72x \\ \frac{d^4y}{dx^4} &= 120x - 72 \\ \frac{d^5y}{dx^5} &= 120 \end{aligned}$$

6.
$$\begin{aligned} g(x) &= x^{\frac{1}{2}} \\ g'(x) &= \tfrac{1}{2}x^{-\frac{1}{2}} \\ g''(x) &= -\tfrac{1}{4}x^{-\frac{3}{2}} \\ &= \frac{-1}{4x^{\frac{3}{2}}} = -\frac{1}{4\sqrt{x^3}} \end{aligned}$$

7.
$$\begin{aligned} g'(x) &= 30x^4 + 8x^3 - 12x^2 + 14x - 8 \\ g''(x) &= 120x^3 + 24x^2 - 24x + 14 \\ g'''(x) &= 360x^2 + 48x - 24 \\ g^{(4)}(x) &= 720x + 48 \\ g^{(5)}(x) &= 720 \\ g^{(6)}(x) &= 0 \\ g^{(7)}(x) &= 0 \end{aligned}$$

Key Idea – Rule 3: Product Rule

1. (a)
$$\begin{aligned} y' &= u'v + v'u \\ &= 1(6x+7) + 6(x-2) \\ &= 6x+7+6x-12 \\ &= 12x-5 \end{aligned}$$
$u = x-2$
$u' = 1$
$v = 6x+7$
$v' = 6$

(b)
$$\begin{aligned} y' &= u'v + v'u \\ &= 3x^2(2x+3) + 2(x^3) \\ &= 6x^3+9x^2+2x^3 \\ &= 8x^3+9x^2 \end{aligned}$$
$u = x^3$
$u' = 3x^2$
$v = 2x+3$
$v' = 2$

(c)
$$\begin{aligned} y' &= u'v + v'u \\ &= 3(5-x) - 1(3x+1) \\ &= 15-3x-3x-1 \\ &= 14-6x \end{aligned}$$
$u = 3x+1$
$u' = 3$
$v = 5-x$
$v' = -1$

(d)
$$\begin{aligned} y' &= u'v + v'u \\ &= 6x(1-2x) - 2(3x^2-1) \\ &= 6x-12x^2-6x^2+2 \\ &= 6x-18x^2+2 \end{aligned}$$
$u = 3x^2-1$
$u' = 6x$
$v = 1-2x$
$v' = -2$

(e)
$$\begin{aligned} y' &= u'v + v'u \\ &= \left(2x-5\right)(2x+3) + 2(x^2-5x) \\ &= 4x^2+6x-10x-15+\ 2x^2-10x \\ &= 6x^2-14x-15 \end{aligned}$$
$u = x^2-5x$
$u' = 2x-5$
$v = 2x+3$
$v' = 2$

(f)
$$\begin{aligned} y' &= u'v + v'u \\ &= -6x(5x^4-x) + \left(20x^3-1\right)(-3x^2) \\ &= -30x^5+6x^2+\left(-60x^5+3x^2\right) \\ &= -30x^5+6x^2-60x^5+3x^2 \\ &= 9x^2-90x^5 \end{aligned}$$
$u = -3x^2$
$u' = -6x$
$v = 5x^4-x$
$v' = 20x^3-1$

(g)
$$\begin{aligned} y &= (5x-8)(2x+3) \\ \frac{dy}{dx} &= vu' + uv' \\ &= 5(2x+3) + 2(5x-8) \\ &= 10x+15+10x-16 \\ &= 20x-1 \end{aligned}$$
$u = 5x-8$
$u' = 5$
$v = 2x+3$
$v' = 2$

(h)
$$\begin{aligned} y &= (x^4-1)(1-x^2) \\ y' &= vu' + uv' \\ &= 4x^3(1-x^2) - 2x(x^4-1) \\ &= 4x^3-4x^5-2x^5+2x \\ &= 4x^3-6x^5+2x \end{aligned}$$
$u = x^4-1$
$u' = 4x^3$
$v = 1-x^2$
$v' = -2x$

(i)
$$\begin{aligned} f(x) &= (x^3+1)(x^2-3x) \\ y' &= vu' + uv' \\ f'(x) &= 3x^2(x^2-3x) + (2x-3)(x^3+1) \\ &= 3x^4-9x^3+2x^4+2x-3x^3-3 \\ &= 5x^4-12x^3+2x-3 \end{aligned}$$
$u = x^3+1$
$u' = 3x^2$
$v = x^2-3x$
$v' = 2x-3$

2. $f(x)=2x(1+x)^3$ Use the product rule as well as the chain rule.

$$\begin{aligned} f'(x) &= vu'+uv' && u=2x \\ &= 2(1+x)^3+2x\times 3(1+x)^2 && u'=2 \\ &= 2(1+x)^3+6x(1+x)^2 && v=(1+x)^3 \\ &= 2(1+x)^2\left[(1+x)+3x\right] && v'=3(1+x)^2 \\ &= 2(1+x)^2\left(1+4x\right) \end{aligned}$$

$f'(0)=2(1+0)^2\left(1+0\right)=2$

3.

$$\begin{aligned} y' &= u'v+v'u && u=3x^2 \\ &= 6x(2x-1)+2(3x^2) && u'=6x \\ &= 12x^2-6x+6x^2 && v=2x-1 \\ &= 18x^2-6x && v'=2 \end{aligned}$$

At $x=2$:

$$\begin{aligned} y' &= 18x^2-6x \\ &= 18(2)^2-6(2) \\ &= 60 \end{aligned}$$

4. Only answer $\dfrac{9\sqrt{2}}{2}$

5. $f'(x)=40x^3-18x^2-10x+3$

$f'(1)=40(1)^3-18(1)^2-10(1)+3$

$=40-18-10+3=15$

6. $\dfrac{-6\pm\sqrt{30}}{3}$

7.

$$\begin{aligned} y &= (2x+3)(x-1) \\ y' &= 2(x-1)+(2x+3) \\ &= 4x+1 \end{aligned}$$

$$\begin{aligned} 4x+1 &= -3 \\ 4x &= -4 \\ x &= -1 \end{aligned}$$

$f(-1)=-2$ Therefore, the point is $(-1,-2)$

<u>Week 3C</u>

Key Idea – Quotient Rule

1. (a)

$$\begin{aligned} y' &= \frac{u'v-v'u}{v^2} && u=3x \\ &= \frac{3\left(x+5\right)-1(3x)}{\left(x+5\right)^2} && u'=3 \\ &= \frac{3x+15-3x}{\left(x+5\right)^2} && v=x+5 \\ &= \frac{15}{\left(x+5\right)^2} && v'=1 \end{aligned}$$

(b)

$$\begin{aligned} y' &= \frac{u'v-v'u}{v^2} && u=x-2 \\ &= \frac{1(5x+7)-5(x-2)}{(5x+7)^2} && u'=1 \\ &= \frac{5x+7-5x+10}{(5x+7)^2} && v=5x+7 \\ &= \frac{17}{(5x+7)^2} && v'=5 \end{aligned}$$

(c)

$$y' = \frac{u'v - v'u}{v^2}$$
$$= \frac{1\left(x^2\right) - 2x(x-5)}{\left(x^2\right)^2}$$
$$= \frac{10x - x^2}{x^4}$$
$$= \frac{x(10-x)}{x^4}$$
$$= \frac{10-x}{x^3}$$

$u = x - 5$

$u' = 1$

$v = x^2$

$v' = 2x$

(d)

$$y = \frac{x-5}{2x+3}$$
$$\frac{dy}{dx} = \frac{vu' - uv'}{v^2}$$
$$= \frac{(2x+3) - 2(x-5)}{(2x+3)^2}$$
$$= \frac{13}{(2x+3)^2}$$

$u = x - 5$

$u' = 1$

$v = 2x + 3$

$v' = 2$

(e)

$$y' = \frac{2x(1-x^2) + 2x(1+x^2)}{(1-x^2)^2}$$
$$= \frac{2x - 2x^3 + 2x + 2x^3}{(1-x^2)^2}$$
$$= \frac{4x}{(1-x^2)^2}$$

$$y' = \frac{vu' - uv'}{v^2}$$

$u = 1 + x^2$

$u' = 2x$

$v = 1 - x^2$

$v' = -2x$

(f)

$$h(x) = \frac{5x^2 - 2x}{2x+5}$$
$$h'(x) = \frac{(2x+5)(10x-2) - 2(5x^2 - 2x)}{(2x+5)^2}$$
$$= \frac{20x^2 - 4x + 50x - 10 - 10x^2 + 4x}{(2x+5)^2}$$
$$= \frac{10x^2 + 50x - 10}{(2x+5)^2}$$

$$y' = \frac{vu' - uv'}{v^2}$$

$u = 5x^2 - 2x$

$u' = 10x - 2$

$v = 2x + 5$

$v' = 2$

2.

$$y' = \frac{u'v - v'u}{v^2}$$
$$= \frac{2x(x-3) - 1(x^2)}{(x-3)^2}$$
$$= \frac{2x^2 - 6x - x^2}{(x-3)^2}$$
$$= \frac{x^2 - 6x}{(x-3)^2}$$

$u = x^2$

$u' = 2x$

$v = x - 3$

$v' = 1$

At $x = 5$:

$$y' = \frac{x^2 - 6x}{(x-3)^2}$$
$$= \frac{5^2 - 6(5)}{(5-3)^2}$$
$$= \frac{-5}{4}$$

Answers only

3. 1/8

4. $x = 0, 1$

5. $x = -9, 3$

Key Idea – Rule 5: Chain Rule

1. Use the chain rule to differentiate.

(a)
$$f(x)=(2x+1)^3$$
$$f'(x)=3(2x+1)^2\times 2$$
$$=6(2x+1)^2$$

(b)
$$y=(x^3-1)^5$$
$$y'=5(x^3-1)^4\times 3x^2$$
$$=15x^2(x^3-1)^4$$

(c)
$$f(x)=(1-x)^4$$
$$f'(x)=4(1-x)^3\times -1$$
$$=-4(1-x)^3$$

(d)
$$y=(x^3-2x^2+5)^3$$
$$\frac{dy}{dx}=3(x^3-2x^2+5)^2\times(3x^2-4x)$$
$$=3(3x^2-4x)(x^3-2x^2+5)^2$$

(e)
$$y=3(7x+4)^7$$
$$\frac{dy}{dx}=21(7x+4)^6\times 7$$
$$=147(7x+4)^6$$

(f)
$$f(x)=\frac{1}{3}(5x+1)^6$$
$$f'(x)=\frac{1}{3}\times 6(5x+1)^5\times 5$$
$$=10(5x+1)^5$$

(g)
$$f(x)=(2-3x)^{-4}$$
$$f'(x)=-4(2-3x)^{-5}\times -3$$
$$=12(2-3x)^{-5}$$
$$=\frac{12}{(2-3x)^5}$$

(h)
$$y=\frac{2}{(x^3+3)^2}=2(x^3+3)^{-2}$$
$$y'=-4(x^2+3)^{-3}\times 3x^2$$
$$=\frac{-12x^2}{(x^3+3)^3}$$

(i)
$$f(x)=-5(x^3-7x^2+x)^{-2}$$
$$f'(x)=-5\times -2(x^3-7x^2+x)^{-3}\times(3x^2-14x+1)$$
$$=10(3x^2-14x+1)(x^3-7x^2+x)^{-3}$$
$$=\frac{10(3x^2-14x+1)}{(x^3-7x^2+x)^3}$$

(j)
$$y=\sqrt{5x-3}=(5x-3)^{\frac{1}{2}}$$
$$y'=\frac{1}{2}(5x-3)^{-\frac{1}{2}}\times 5$$
$$=\frac{5}{2}(5x-3)^{-\frac{1}{2}}$$
$$=\frac{5}{2\sqrt{5x-3}}$$

(k)
$$y=\frac{6}{\sqrt{4+x}}=6(4+x)^{-1/2}$$
$$y'=6\times -\frac{1}{2}(4+x)^{-\frac{3}{2}}$$
$$=\frac{-3}{\sqrt{(4+x)^3}}$$

(l)
$$y=\frac{2}{(4x-1)^{\frac{1}{5}}}=2(4x-1)^{\frac{-1}{5}}$$
$$y'=2\times\frac{-1}{5}(4x-1)^{\frac{-6}{5}}\times 4$$
$$y'=\frac{-8}{5}(4x-1)^{\frac{-6}{5}}$$
$$y'=\frac{-8}{5(4x-1)^{\frac{6}{5}}}\ \text{or}\ \frac{-8}{5\sqrt[5]{(4x-1)^6}}$$

2.
$$f(x)=\frac{(x^2+1)^4}{2}$$
$$f'(x)=\tfrac{1}{2}\times 4(x^2+1)^3\times 2x$$
$$=4x(x^2+1)^3$$
$$f'(1)=4\times(1+1)^3=32$$

3. $f(x) = 3 + 4(2x-1)^3$

$$f'(x) = 0 + 4 \times 3(2x-1)^2 \times 2$$
$$= 24(2x-1)^2$$

$Solve: f'(x) = 24(2x-1)^2 = 0$

$$24(2x-1)^2 = 0$$
$$2x - 1 = 0$$
$$x = \frac{1}{2}$$

4.

$$f(x) = 5(1-x)^3$$
$$f'(x) = -15(1-x)^2$$
$$f'(-1) = -15\left(1-(-1)\right)^2$$
$$= -60$$

Key Idea – Derivative of Powers of *x* Combining Rule 1-5

Answer only

1. (a) $f'(x) = 10x^4(8x+3)(5x+3)^2$ (b) $y'(x) = \frac{-1}{\sqrt{5-2x}}$

2. -2

3. (a) $y' = \frac{-17x^6 + 340x^3 - 9x^2}{(x^4-3)^6}$ (b) $y' = \frac{x-2}{2\left(x-1\right)^{3/2}}$

<u>Week 3D</u>

Key Idea – Further Applications of Derivatives - Equation of Tangent

1. (a)

$$y = x^2$$
$$y' = 2x$$
$$y'(2) = 4$$
$$y - y_1 = m(x - x_1)$$
$$y - 4 = 4(x-2)$$
$$y - 4 = 4x - 8$$
$$4x - y - 4 = 0$$

(b)

$$y = x^2 - 5x + 6$$

Let $x = 0$

$$y = 0^2 - 5 \times 0 + 6$$
$$y = 6$$

The point is (0, 6)

$$y' = 2x - 5$$
$$y'(0) = 2 \times 0 - 5 = -5$$
$$y - y_1 = m(x - x_1)$$
$$y - 6 = -5(x - 0)$$
$$y = -5x + 6$$
$$5x - 6 = 0$$

(c)

$$y = 3x^3 - 7x^2 + 2x$$
$$y' = 9x^2 - 14x + 2$$
$$y(2) = 3(2)^3 - 7(2)^2 + 2(2)$$
$$= 0$$
$$\therefore (2, 0)$$

(d)

$$y = 2x^2 - 4x + 1$$
$$\frac{dy}{dx} = 4x - 4$$
$$\therefore 4x - 4 = 4$$
$$4x = 8$$
$$x = 2$$

$$y'(2) = 9(2)^2 - 14(2) + 2$$
$$= 10$$
$$y - y_1 = m(x - x_1)$$
$$y - 0 = 10(x - 2)$$
$$y = 10x - 20$$
$$10x - y - 20 = 0$$

$$y(2) = 2(2)^2 - 4(2) + 1$$
$$= 1$$
$$\therefore (2,1) \text{ when } m = 4$$
$$y - 1 = 4(x - 2)$$
$$= 4x - 8$$
$$4x - y - 7 = 0$$

(e) $y = x^2 - x - 12$

Let $y = 0$

$$(x-4)(x+3) = 0$$
$$x = 4,\ x = -3$$

The points are (4,0) and (−3,0)

Equation 1 at (4,0)

$$y' = 2x - 1$$
$$y'(4) = 7$$
$$y - 0 = 7(x - 4)$$
$$y = 7x - 28$$
$$7x - y - 28 = 0$$

Equation 2 at (−3,0)

$$y' = 2x - 1$$
$$y'(-3) = -7$$
$$y - 0 = -7(x + 3)$$
$$y = -7x - 21$$
$$7x + y + 21 = 0$$

(f) Answer only

$y = 4x - 6$

Key Idea – Derivatives of Exponential Functions

1. (a) $y' = 4e^{4x}$ (b) $\dfrac{dy}{dx} = -e^{-x}$ (c) $y' = 4e^{4x+1}$

(d) $y' = -1 \times (2x-4)e^{x^2-4x+1}$
$= (-2x+4)e^{x^2-4x+1}$

(e) $y' = 2 \times 12x^3 e^{3x^4}$
$= 24x^3 e^{3x^4}$

(f) $y' = \dfrac{1}{2} \times 6e^{6x}$
$= 3e^{6x}$

(g) $y' = 6x^2 + 5e^{5x}$ (h) $y' = 6x - e^{1-x}$ (i) $f(x) = e^{2x}\left(e^x - e^{-x}\right) = e^{3x} - e^x$
$f'(x) = 3e^{3x} - e^x$

2. (a)
$$y = 2x^3 e^{5x}$$
$$y' = vu' + uv'$$
$$y' = e^{5x} \times 6x^2 + 2x^3 \times 5e^{5x}$$
$$= 6x^2 e^{5x} + 10x^3 e^{5x}$$

$$u = 2x^3$$
$$u' = 6x^2$$
$$v = e^{5x}$$
$$v' = 5e^{5x}$$

(b)
$$y = \frac{e^{2x} + 1}{2x - 1}$$
$$y' = \frac{(2x-1)2e^{2x} - \left(e^{2x}+1\right)2}{(2x-1)^2}$$
$$= \frac{4xe^{2x} - 2e^{2x} - 2e^{2x} - 2}{(2x-1)^2}$$
$$= \frac{4xe^{2x} - 4e^{2x} - 2}{(2x-1)^2}$$

$$y' = \frac{vu' - uv'}{v^2}$$
$$u = e^{2x} + 1$$
$$u' = 2e^{2x}$$
$$v = 2x - 1$$
$$v' = 2$$

Answers only

(c) $f'(x) = 18\left(2e^x - 1\right)^5 \times 2e^x$
$= 36e^x\left(2e^x - 1\right)^5$

(d) $f'(x) = 3(1+2e^{2x})\left(x + e^{2x}\right)^2$

(e) $f'(x) = x^2(3+2x)e^{2x-1}$

(f) $y' = \dfrac{4e^{2x+1}(x+2)}{(2x+5)^2}$

3.

(a) $g(x) = 4e^{2x-1}$

$g'(x) = 4 \times 2e^{2x-1}$

$= 8e^{2x-1}$

$g'(2) = 8e^3$

(b) $y = e^{2x} + e^{8x}$

$\dfrac{dy}{dx} = 2e^{2x} + 8e^{8x}$

$\dfrac{d^2y}{dx^2} = 4e^{2x} + 64e^{8x}$

(c) $f\left(x\right) = -2e^{3x}$

$f\left(1\right) = -2e^3$

$\therefore$ Point is $(1, -2e^3)$

$f'\left(x\right) = -6e^{3x}$

$f'\left(1\right) = -6e^3$

$\therefore$ Slope is $-6e^3$

$y - y_1 = m(x - x_1)$

$y + 2e^3 = -6e^3(x-1)$

Equation: $6e^3x + y - 4e^3 = 0$

(e) $f\left(x\right) = 2xe^x$ $\quad u = 2x$, $u' = 2$, $v = e^x$, $v' = e^x$

$y' = vu' + uv'$

$f'\left(x\right) = 2e^x + 2xe^x$

$f'\left(x\right) = 0$ when $2e^x\left(1+x\right) = 0 \quad x = -1$

$f\left(-1\right) = 2 \times -1 \times e^{-1} = -2e^{-1}$ $\quad \therefore$ Gradient equals zero at $\left(-1, -\dfrac{2}{e}\right)$

4.

$y = e^{x^3+1}$

$y' = 3x^2e^{x^3+1}$

$y'(-1) = 3 \times 1 \times e^0 = 3$

$y(-1) = e^0 = 1$

Point $(-1, 1)$ At $x = -1$, $m = 3$

$y - y_1 = m(x - x_1)$

$y - 1 = 3(x+1)$

$y - 1 = 3x + 3$

$3x - y + 4 = 0$